

Caesar Cipher Practical Exercise

Background

The Caesar cipher is one of the simplest encryption techniques, where each letter in the plaintext is shifted a certain number of positions down the alphabet. For example, with a shift of 3, 'A' becomes 'D', 'B' becomes 'E', and so on.

Task 1: Basic Encryption

Instructions

1. Using the Caesar cipher wheel or cryptii.com, encrypt the following message with a shift of 3:
`SECURITY IS IMPORTANT`
2. Write down your process and the encrypted result
3. Share your result with a classmate and verify if you got the same output

Task 2: Decryption Challenge

Instructions

1. The following message was encrypted using a Caesar cipher. Your task is to decrypt it:
`ZHOFRPH WR FUBSWRJUDSKB`
2. Hints:
 - The shift value is between 1 and 5
 - The decrypted message makes sense in English
 - First word might be "WELCOME"

Task 3: Breaking the Cipher

Instructions

1. Without knowing the shift value, decrypt this message:
`PBVHLX TL MAX YBKLM LMXI`
2. Method:
 - Try each possible shift value (1-25)
 - Look for English words
 - Record each attempt until you find readable text

Assessment Questions

1. Why does the Caesar cipher wrap around the alphabet?
2. How many possible shift values are there in a Caesar cipher?
3. What makes the Caesar cipher easy to break?
4. Can you think of ways to make the Caesar cipher more secure?

This exercise helps students understand: - Basic encryption and decryption concepts - The importance of key selection (shift value) - Weaknesses in simple substitution ciphers - The relationship between plaintext and ciphertext